RP2350pc

User Manual

Rev.1.0 June 2025

olimex.com

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What is RP2350pc

RP2350pc is complete all in one computer based on RP2350 Dual core Cortex-M33 + Dual core Hazard3 RISC-V processor from the Raspberry Pi foundation.

The features of RP2350pc are:

- RP2350B SOC with easy to load new firmware via drag and drop virtual drive
- 520 KB on-chip SRAM
- 16MB SPI Flash
- 8MB of PSRAM
- DVI/HDMI output
- USB hub with x4 USB2.0 hosts which can be used to connect to keyboard, mouse, USB Flash, USB Gamepads etc
- Stereo Audio codec
- Stereo Amplifier
- Audio 3.5mm connector Line In
- Audio 3.5mm connector for Headphones
- JST2.0 connectors for Left and Right speakers
- USB-C connector for power supply
- USB-C connector for programming
- Two UEXT connectors with I2C, UART and SPI for connecting to external boards
- Power switch
- Reset and Boot buttons
- four mounting holes 3.3mm diameter
- Lipo battery charger which allow the board to run from LiPo battery.
- Lipo JST2.0 mm connector
- Dimension 85x65mm

RP2350pc is Open Source Hardware, all CAD files and firmware and available, so people can study and modify.

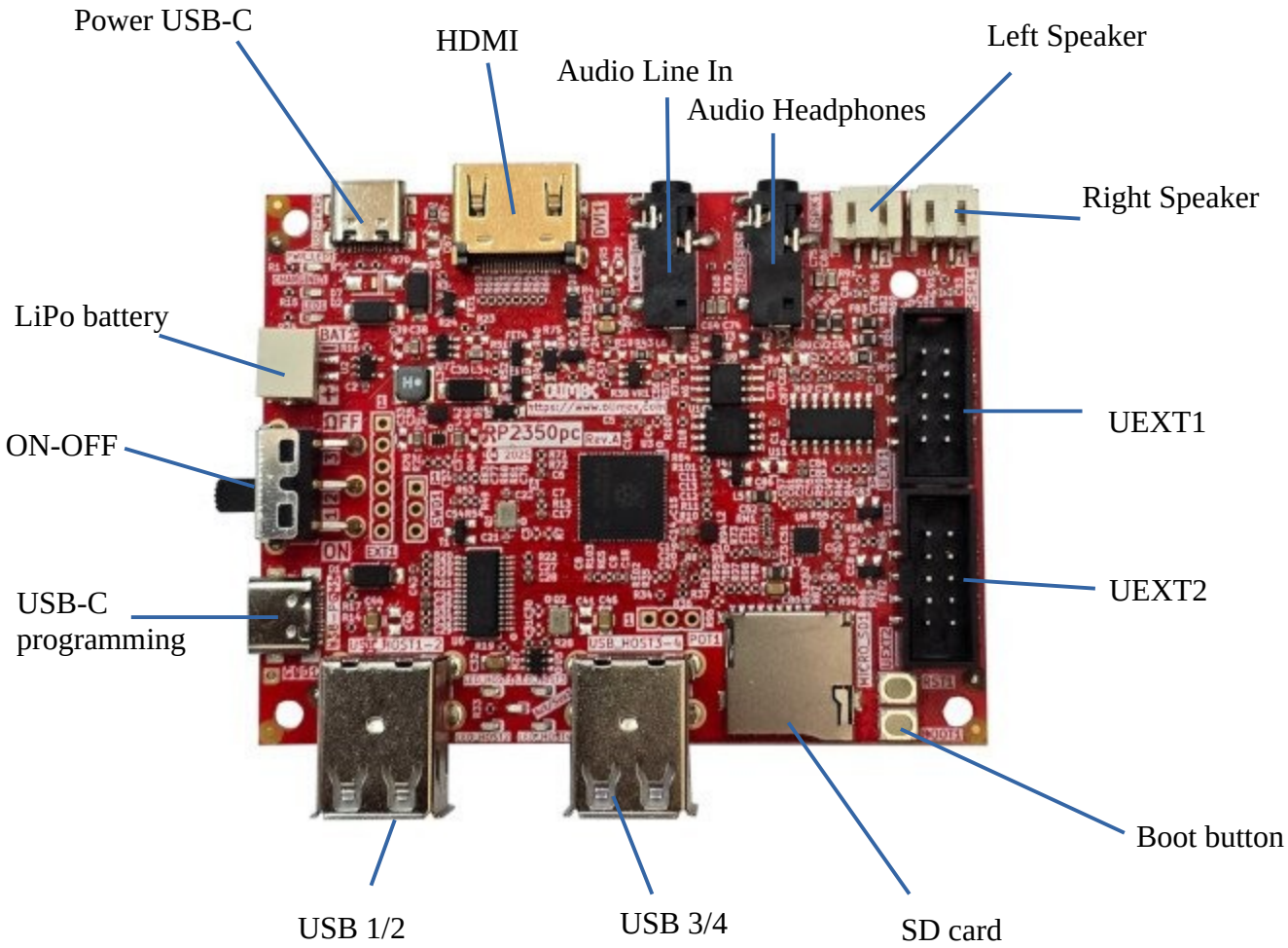
Important notice: If RP2350pc is not mounted in box be careful to not place it on metal surface, nor drop metal objects on top of the PCB! This will lead to damage.

Order codes for RP2350pc and accessories:

RP2350pc	RP2350 all in one computer with 4 USB hosts and HDMI display
USB-KEYBOARD-PS2	Keyboard which is compatible with RP2350pc
USB-GAMEPAD	USB Gamepad
USB-WIRELESS-GAMEPAD	USB Wireless Gamepad
USB-CABLE-AM-USB3-C	High speed, High current cable for power supply and programming
CABLE-HDMI-50CM	HDMI cable
UEXT modules	many UEXT modules which can connect to Neo6502 UEXT connector
BATTERY-LiPo1400mAh	LiPo battery compatible with RP2350pc

HARDWARE

RP2350pc layout:

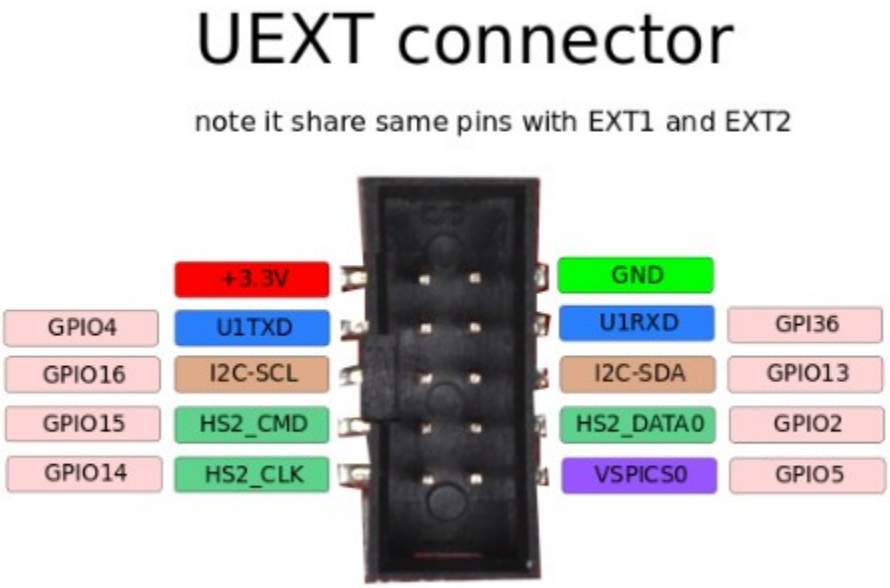


UEXT connector:

UEXT connector stands for Universal EXTension connector and contain +3.3V, GND, I2C, SPI, UART signals.

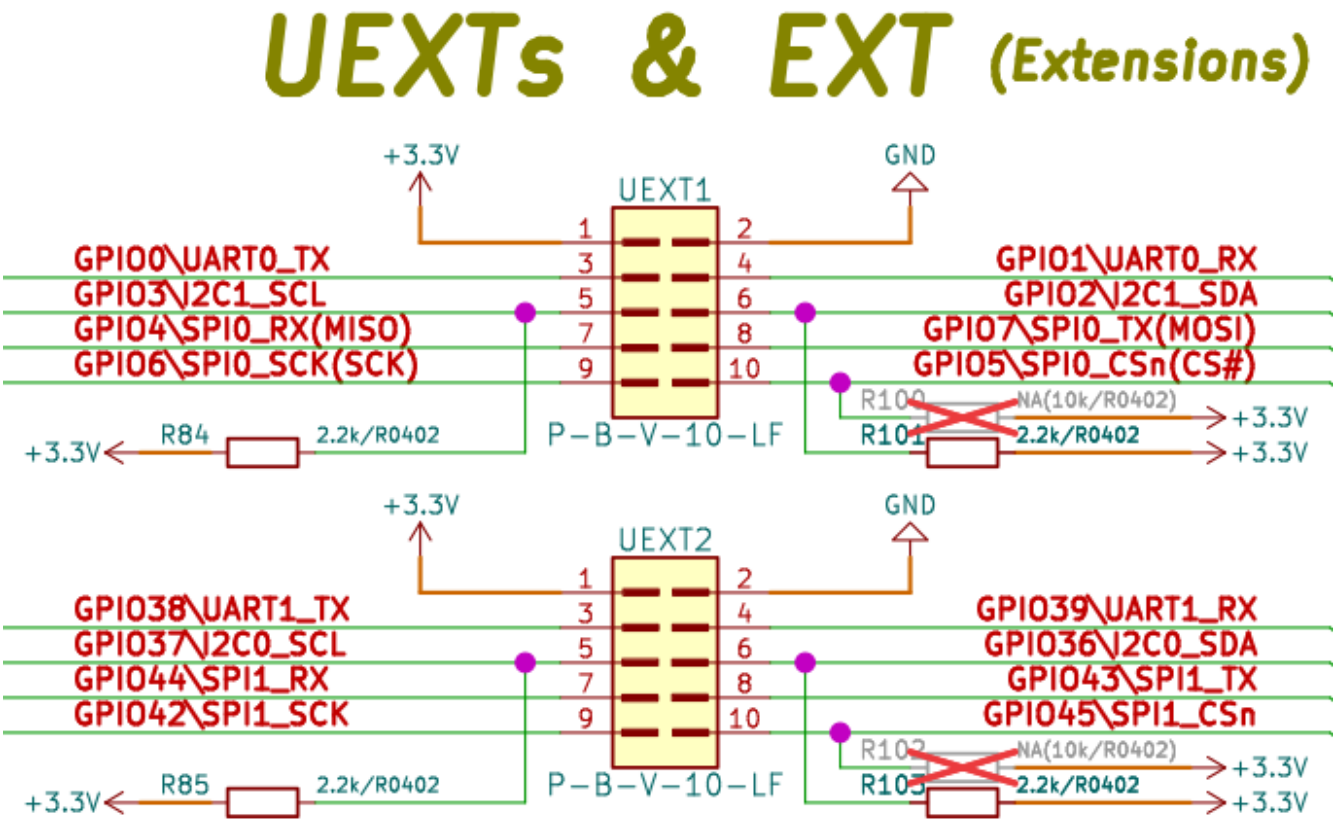
UEXT connector can be in different shapes.

The original UEXT connector is 0.1” 2.54mm step boxed plastic connector. All signals are with 3.3V levels.

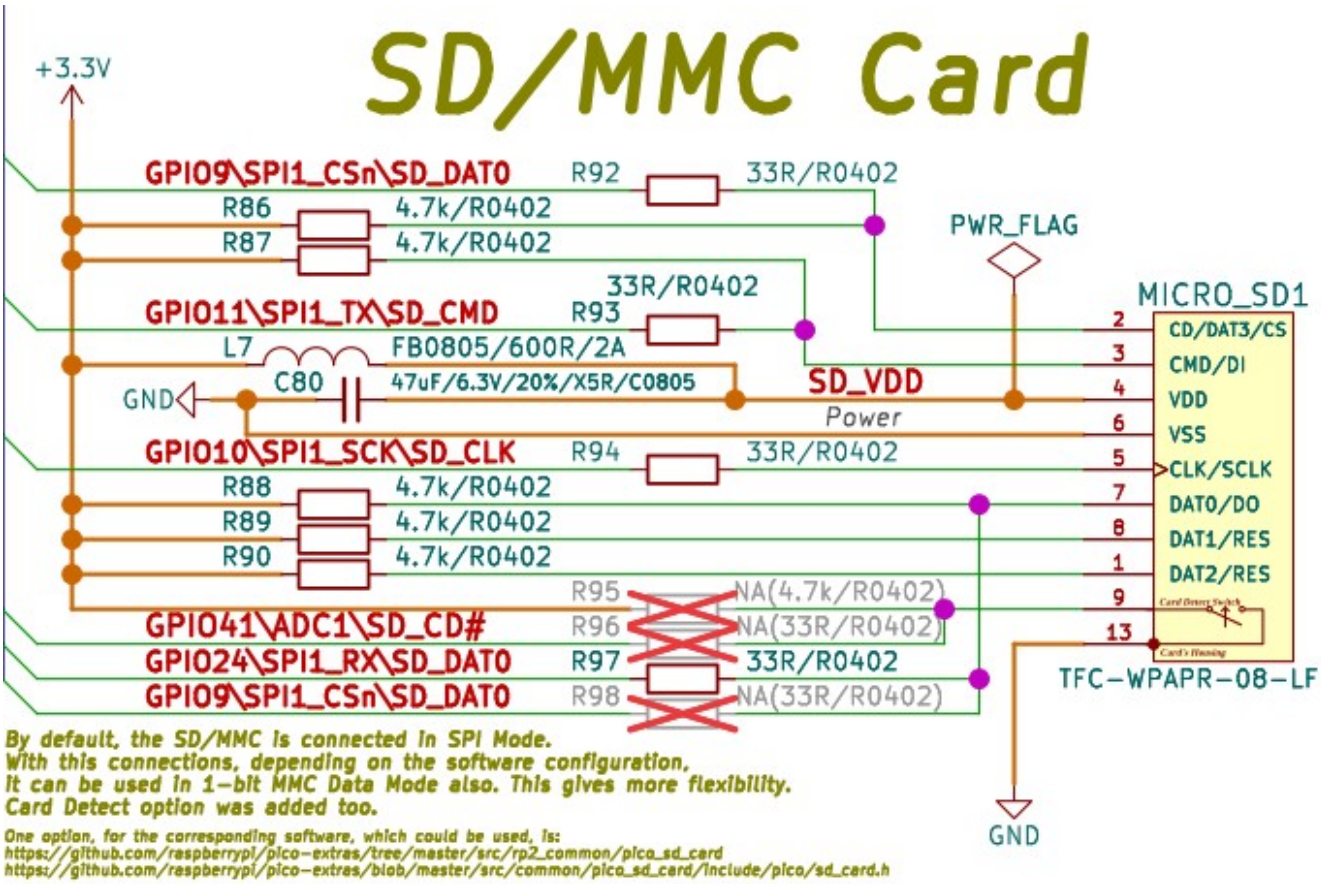


Olimex has developed number of MODULES with this connector. There are temperature, humidity, pressure, magnetic field, light sensors. Modules with LCDs, LED matrix, Relays, Bluetooth, Zigbee, WiFi, GSM, GPS, RFID, RTC, EKG, sensors and etc.

RP2350pc UEXT connectors:

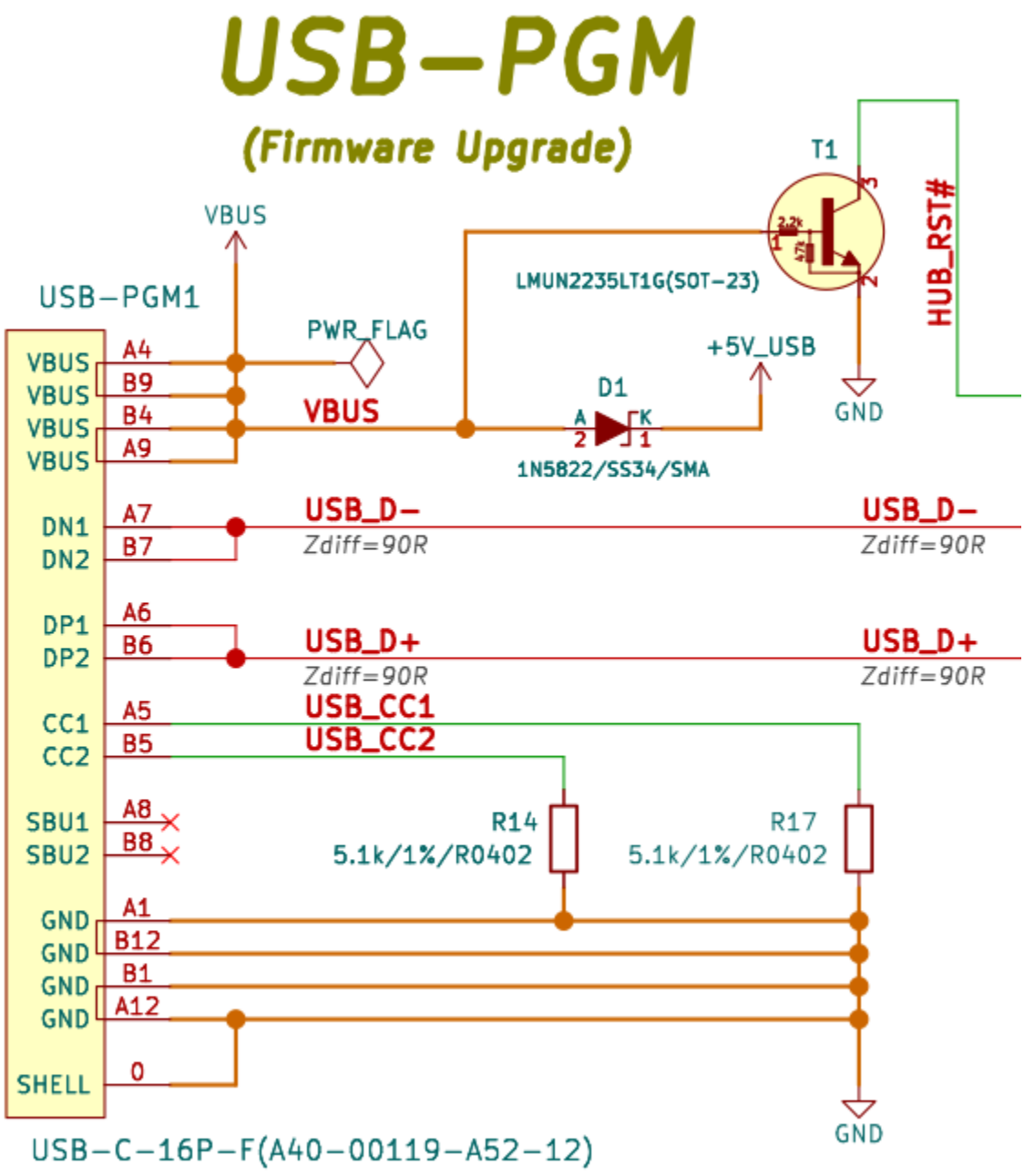


SD-card interface:



USB-C programming connector:

It automatically disables the USB-HUB just press the boot button insert USB-C cable and the RP2350 go in bootloader mode and make disk.



RP2350pc schematics:

RP2350pc latest schematic is on GitHub

SOFTWARE:

RP2350pc can be programmed with RaspberryPi C-SDK or MicroPython SDK.

For the retro computing fans the Reload emulator written by Veselin Sladkov will support RP2350pc soon and will emulate Apple][, Apple][e, Oric Atmos, Pravetz 82, Pravetz 8D and all games from Total Replay 5.2 are supported.

Paul Robson works on RP2350pc API which will allow compilers and OS to be created with unified API (BIOS).

Programming RP2350pc

The RP2350 firmware is UF2 file. You will be able to get pre-build firmware of reload emulator on olimex’s ftp when available.

To program the .uf2 files you need USB-A to USB-C cable like USB-CABLE-AM-USB3-C.

1. Disconnect the power supply from USB-PWR1 connector and connect it to USB-PGM1 connector.
2. Press the BOOT1 button and switch on the power supply with PWR_ON/OFF1 switch then release BOOT1 button.
3. You will see on your computer new disk drive RPI-RP2.
4. Copy the .uf2 file to this drive, once it’s copied the drive will disappear.
5. Switch OFF the PWR_ON/OFF1 switch
6. Disconnect the USB-C cable from USB-PGM1 and connect to USB-PWR1 connector.
7. Switch ON power supply.

Revision History

Revision 1.0 June 2025